

EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science in Electrical Engineering

Minor in Computer Science & Semiconductor Engineering

GPA 3.57/4.0

August 2024 – May 2028

SKILLS & RELEVANT COURSE WORK

- **Languages:** C, C++, Python, SystemVerilog, RISC-V Assembly, VB.NET
- **Protocols:** CAN, LIN
- **OS:** Linux, Windows
- **Tools:** Xilinx Vivado, LTspice, KiCad, Git
- **Lab Equipment:** Oscilloscope, Function Generator, Power Supply, Digital Multimeter, LabJack
- **Relevant Coursework (See website for details):** Data Structures, Computer Systems, Digital Systems, Analog Signal Processing, Power Circuits & Electromechanics

EXPERIENCE

Micro-Nano Mechanical Systems (MNMS) Lab Technician

University of Illinois Urbana-Champaign | Urbana, Illinois

September 2026 - Present

- Support cleanroom laboratory operations, equipment setup, maintenance, and technical processes for micro- and nanoscale mechanical systems research.

Vehicle Solutions Engineering Intern

Grayhill, Inc. | La Grange, Illinois

May 2026 – August 2026

- Developed hardware test systems for CAN/LIN keypads, joysticks, and encoders, integrating LabJack DAQ and communication interfaces for device validation
- Designed and simulated test circuitry in LTspice, selecting surface-mount components based on voltage, current, and power requirements
- Performed hardware verification and validation testing, troubleshooting electrical interfaces and developing test procedures and technical documentation

Testing Engineering Intern

Teinnovations LLC | Aurora, Illinois

May 2025 – August 2025

- Developed and prepared test samples using lab protocols to enable accurate and repeatable testing
- Streamlined testing pipeline by performing 500+ standardized tests on polymer films (PP/PET), providing critical data for material performance reports
- Analyzed test results to identify performance trends and support material validation

PROJECTS

16-Bit Adder Architecture

September 2026

- Designed and implemented 16-bit ripple-carry, carry-lookahead, and carry-select adders in SystemVerilog.
- Built a hierarchical 4×4 carry-lookahead architecture using propagate/generate logic and group carry equations to reduce carry propagation delay.
- Developed SystemVerilog testbenches to verify arithmetic functionality and debug logic prior to hardware implementation.

Bit-Serial Processor

August 2026

- Designed and implemented a 4-bit bit-serial processor using shift registers, multiplexers, flip-flops, combinational logic, and an FSM-based control unit.
- Used provided SystemVerilog testbenches to simulate and verify datapath and control behavior before hardware implementation.
- Synthesized the design in Xilinx Vivado and used the Integrated Logic Analyzer (ILA) debug core to inspect internal signals and verify FPGA behavior.

ACTIVITIES

Electrical Lead

The Boring Illini

July 2026 – Present

- Leading development of the electrical architecture for a tunnel boring machine, including power distribution, actuator control, motor control, sensors, embedded systems, and system integration
- Secured actuator sponsorship and coordinated component selection and integration planning for the team's electrical systems